



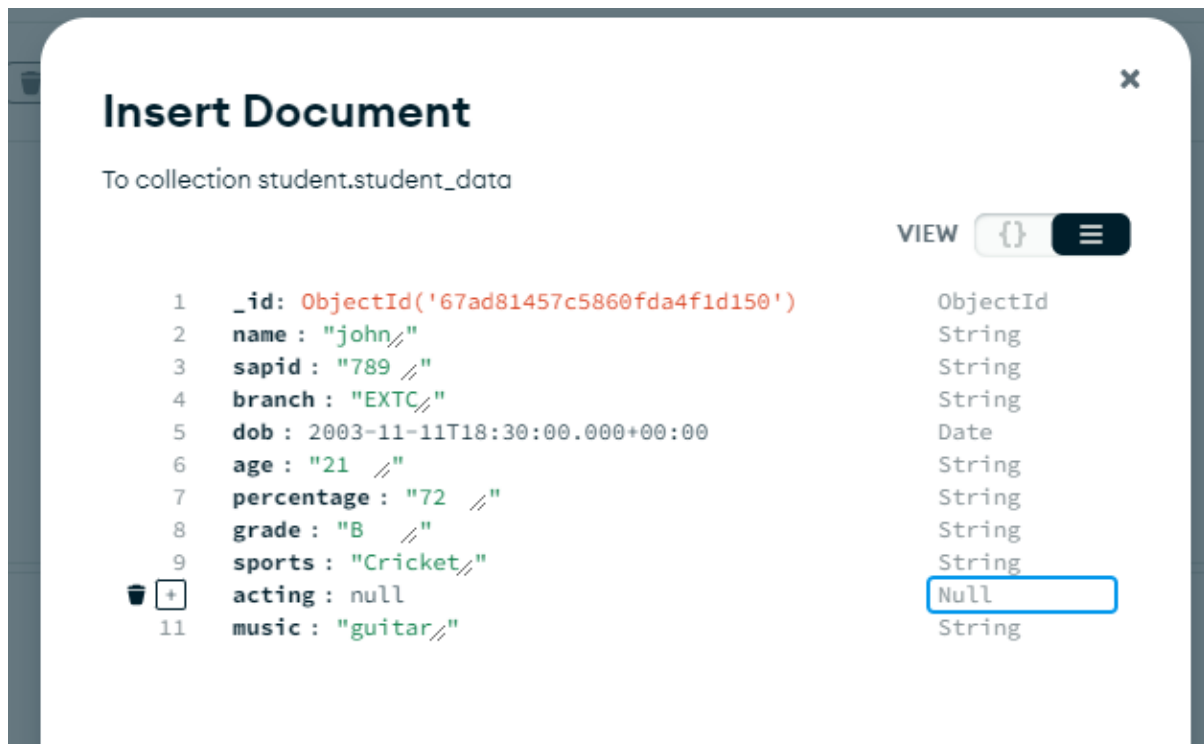
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EXPERIMENT 4

Aim: Advanced querying and data manipulation in MongoDB.

Screenshots:



The screenshot shows the MongoDB 'Insert Document' interface. The title is 'Insert Document' with a close button (X) in the top right. Below the title, it says 'To collection student.student_data'. On the right, there is a 'VIEW' button and a toggle switch set to 'JSON' (represented by curly braces). The main area displays a document being inserted, with line numbers 1 through 11 on the left. The document fields are: `_id` (ObjectId), `name` (String), `sapid` (String), `branch` (String), `dob` (Date), `age` (String), `percentage` (String), `grade` (String), `sports` (String), `acting` (null), and `music` (String). On the right side, the data types for each field are listed: ObjectId, String, String, String, Date, String, String, String, String, and String. The 'acting' field is highlighted with a blue box, and its value 'null' is also highlighted with a blue box.

```
1  _id: ObjectId('67ad81457c5860fda4f1d150')
2  name : "john/"
3  sapid : "789 //"
4  branch : "EXTC/"
5  dob : 2003-11-11T18:30:00.000+00:00
6  age : "21 //"
7  percentage : "72 //"
8  grade : "B //"
9  sports : "Cricket/"
10 acting : null
11 music : "guitar/"
```

VIEW ☒ ☐

ObjectId
String
String
String
Date
String
String
String
String
String
Null
String



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+ ADD DATA EXPORT DATA UPDATE DELETE

```
_id: ObjectId('67ad7e977c5860fda4f1d14a')
name : "abc"
sapid : "123"
branch : "IT"
dob : 2004-11-30T18:30:00.000+00:00
age : "20"
percentage : 92
grade : "O"
sports : "football"
acting : "theatre"
music : "piano"
```

```
_id: ObjectId('67ad7f1f7c5860fda4f1d14c')
name : "pqr"
sapid : "234"
branch : "CSE(ICB)"
dob : 2002-02-10T18:30:00.000+00:00
age : 23
percentage : 94
grade : "O"
sports : null
acting : "street plays"
music : "singing"
```

```
_id: ObjectId('67ad7f987c5860fda4f1d14d')
name : "uvw"
sapid : "567"
branch : "Comps"
dob : 2005-05-03T18:30:00.000+00:00
age : 19
percentage : 85
grade : "A"
sports : null
acting : null
music : null
```

Aggregation:

1) Match

Stage1 \$match

```
1 // **
2 * Provide the field name for the count.
3 */
4 {'percentage': {'$gte': 90}}
```

Output after \$match stage (Sample of 2 documents)

```
_id: ObjectId('67ad7e977c5860fda4f1d14a')
name : "abc"
sapid : "123"
branch : "IT"
dob : 2004-11-30T18:30:00.000+00:00
age : "20"
percentage : 92
grade : "O"
sports : "football"

_id: ObjectId('67ad7f1f7c5860fda4f1d14c')
name : "pqr"
sapid : "234"
branch : "CSE(ICB)"
dob : 2002-02-10T18:30:00.000+00:00
age : 23
percentage : 94
grade : "O"
sports : null
```



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2) Group

Stage 1 \$group

```
1 /**
2  * Provide the field name for the count.
3  */
4  { "_id": "$grade",
5    "avgPercentage": { "$avg": "$percentage" }
6  }
```

Output after \$group stage (Sample of 2 documents)

```
_id: "O"
avgPercentage: 93
```

```
_id: "A"
avgPercentage: 85
```

3) Project

Stage 1 \$match

```
1 /**
2  * Provide the field name for the count.
3  */
4  { "percentage": { "$gte": 75 } }
5  }
```

Output after \$match stage (Sample of 3 documents)

```
sapId: "123"
branch: "IT"
dob: 2004-11-30T18:30:00.000+00:00
age: 20
percentage: 92
grade: "O"
sports: "football"
acting: "theatre"
music: "piano"
```

```
_id: ObjectId('67ad7f1f7c5860fda4f1d14c')
name: "pqr"
sapId: "234"
branch: "CSE(ICB)"
dob: 2002-02-10T18:30:00.000+00:00
age: 23
percentage: 94
grade: "O"
sports: null
```

```
_id: ObjectId('67ad7f987c5860fda4f1d14d')
name: "uvu"
sapId: "567"
branch: "Comps"
dob: 2005-05-03T18:30:00.000+00:00
age: 19
percentage: 85
grade: "A"
sports: null
```

Stage 2 \$group

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5  {
6    "_id": "$grade", // Group by 'grade'
7    "averagePercentage": { "$avg": "$perc
8  }
9  }
```

Output after \$group stage (Sample of 2 documents)

```
_id: "A"
averagePercentage: 85
```

```
_id: "O"
averagePercentage: 93
```

Stage 3 \$project

```
1 /**
2  * specifications: The fields to
3  * include or exclude.
4  */
5  {
6    "_id": 0, // Exclude the _id
7    "grade": "$_id", // Rename _id to
8    "averagePercentage": 1 // Keep averag
9  }
10
```

Output after \$project stage (Sample of 2 documents)

```
averagePercentage: 93
grade: "O"
```

```
averagePercentage: 85
grade: "A"
```

+ Add Stage

Pull Operator:



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The screenshot shows the MongoDB Compass interface. On the left, a document is displayed with the following fields: `_id`, `name`, `sapid`, `branch`, `dob`, `age`, `percentage`, `grade`, `sports`, `acting`, and `music`. The `music` field is an array containing `"singing"`, `"piano"`, and `"guitar"`. In the center, an 'Update 1 document' dialog box is open. It shows the filter `{ percentage: { $gte: 93 } }` and the update operation `{ $pull: { acting: "Street plays" } }`. The dialog has buttons for 'Save', 'Cancel', and 'Update 1 document'.

```
_id: ObjectId('67ad7f1f7c5860fda4f1d14c')
name: "pqr"
sapid: "234"
branch: "CSE(ICB)"
dob: 2002-02-10T18:30:00.000+00:00
age: 23
percentage: 94
grade: "O"
sports: null
acting: "street plays"
music: Array (3)
  0: "singing"
  1: "piano"
  2: "guitar"
```



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Update 1 document

student.student_data

Filter ⓘ

{ percentage: { \$gte: 93 } }

Update

[Learn more about Update syntax](#) ⓘ

```
1 {
2   $pull: {
3     "music": "singing"
4   },
5 }
```

★ Save

Cancel

Update 1 document

```
_id: ObjectId('67ad7f1f7c5860fda4f1d14c')
name : "pqr"
sapid : "234"
branch : "CSE(ICB)"
dob : 2002-02-10T18:30:00.000+00:00
age : 23
percentage : 94
grade : "0"
sports : null
acting : "street plays"
▼ music : Array (2)
  0: "piano"
  1: "guitar"
```

Conclusion: Thus we successfully implemented advanced querying and data manipulation in MongoDB.